

Chapter 2. PERMUTATIONS

2.1 What is a permutation?



Today we will introduce a new object, called

PERMUTATIONS

2.1 What is a permutation?

From a mathematical point of view

A **permutation** of a set X is a bijection $\alpha : X \rightarrow X$. Given an integer $n \geq 1$, write $X_n = \{1, 2, \dots, n\}$ then the set of all permutations of X_n is denoted by S_n , and is called the **symmetric group of degree n** .

Does it correspond to our intuitive idea about permutations?

Let us consider the numbers 1, 2, 3. A permutation of 1, 2, 3 is a rearrangement of them in a definite order. We have the six following possibilities:

1 2 3 1 3 2 2 1 3 2 3 1 3 1 2 3 2 1

Let us now look at our definition of permutations of $\{1, 2, 3\}$:

Set $X_3 = \{1, 2, 3\}$ and think now about the bijective maps $X_3 \rightarrow X_3$:

$$\begin{array}{l} 1 \mapsto 1 \\ 2 \mapsto 2 \\ 3 \mapsto 3 \end{array} \quad \left| \quad \begin{array}{l} 1 \mapsto 1 \\ 2 \mapsto 3 \\ 3 \mapsto 2 \end{array} \right| \quad \left| \quad \begin{array}{l} 1 \mapsto 2 \\ 2 \mapsto 1 \\ 3 \mapsto 3 \end{array} \right| \quad \left| \quad \begin{array}{l} 1 \mapsto 2 \\ 2 \mapsto 3 \\ 3 \mapsto 1 \end{array} \right| \quad \left| \quad \begin{array}{l} 1 \mapsto 3 \\ 2 \mapsto 1 \\ 3 \mapsto 2 \end{array} \right| \quad \left| \quad \begin{array}{l} 1 \mapsto 3 \\ 2 \mapsto 2 \\ 3 \mapsto 1 \end{array} \right|$$

Same results written in two different ways!

We write down permutations using the Matrix-Type Notation.

MATRIX-TYPE NOTATION. We write $\sigma \in S_n$ as a matrix as follows:

- in the top row we list the elements $1, 2, \dots, n$ of X_n ;
- in the bottom row we put the numbers after shuffling.

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix} = \begin{pmatrix} 1 & 2 & \dots & n \\ \downarrow & \downarrow & & \downarrow \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix}$$

In other words, the elements of X_n are listed in the top row, and beneath each element of X_n is its image.

Example 2.1. The permutation $\sigma \in S_4$ defined by $\sigma(1) = 3$, $\sigma(2) = 1$, $\sigma(3) = 4$, $\sigma(4) = 2$ in matrix notation is:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}$$

The permutation $\beta \in S_6$ given by $\beta(1) = 6$, $\beta(2) = 3$, $\beta(3) = 1$, $\beta(4) = 5$, $\beta(5) = 2$ and $\beta(6) = 4$ is expressed in matrix notation as

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 3 & 1 & 5 & 2 & 4 \end{pmatrix}$$

THEOREM 2.2. The symmetric group S_n of degree n has $|S_n| = n!$ elements.

Proof. Recall that the elements of S_n are bijective maps $\underbrace{\{1, 2, \dots, n\}}_{\bar{X}_n} \longrightarrow \{1, 2, \dots, n\}$

Two permutations σ and τ in S_n are equal if and only if $\sigma(k) = \tau(k)$ for all $k = 1, \dots, n$.

To construct a permutation σ we have to choose $\sigma(1), \sigma(2), \dots, \sigma(n)$ in $\bar{X}_n$ such that they are all different.

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \dots & \sigma(n) \end{pmatrix}$$

For $\sigma(1)$ we have n choices (any element of X_n); for $\sigma(2)$ we have $n-1$ choices (elements of $X_n \setminus \{\sigma(1)\}$), for $\sigma(3)$ we have $n-2$ choices (elements of $X_n \setminus \{\sigma(1), \sigma(2)\}$).

In total we have

$$n(n-1)(n-2) \dots 2 \cdot 1 = n!$$

permutations.

